3.6 Using the Quadratic Formula

Name _____ Period ____ Date _____

Algebra II 3.6 · Student sheet

I can…

- solve quadratic equations using the Quadratic Formula.
- use the discriminant to find the number and type of solutions.
- use the Quadratic Formula to solve real-life problems.

Vocabulary · fill in the missing word

Quadratic Formula — $x =$ _____, for $ax^2 + bx + c = 0$

Discriminant — _____

$b^2 - 4ac > 0$ — two real _____

$b^2 - 4ac = 0$ — one real _____

$b^2 - 4ac < 0$ — two _____ solutions

Part 1 · Discriminant sort

Equation	$b^2 - 4ac$	Predicted x-intercepts	Graph agrees?
$x^2 - 4x + 4 = 0$			
$x^2 + 2x + 5 = 0$			
$x^2 - x - 6 = 0$			
$2x^2 + 3x + 1 = 0$			
$x^2 + 6x + 9 = 0$			
$3x^2 - x + 2 = 0$			

Part 2 · Notes: fill in the blanks

$x =$ _____

Discriminant _____ > 0: _____; = 0: _____; < 0: _____.

Part 3 · Examples (work them with the class)

Example 1

Solve $x^2 + 3x - 5 = 0$ using the Quadratic Formula.

Example 2

Solve $2x^2 - 4x + 5 = 0$.

Example 3

Find the discriminant and describe the solutions.

a. $4x^2 - 12x + 9 = 0$ b. $x^2 + 2x + 7 = 0$ c. $3x^2 + x - 2 = 0$

Example 4

A ball is thrown upward. Its height h (in feet) after t seconds is $h = -16t^2 + 30t + 6$. When does it hit the ground?

Part 4 · Partner practice

Partner 1

Solve $x^2 - 6x + 4 = 0$.

Partner 2

Find the number and type of solutions of $5x^2 + 2x + 1 = 0$.